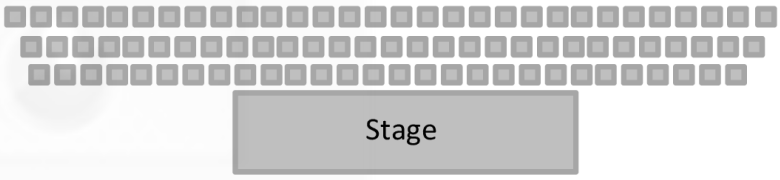
Question 7 **(55 marks)**

Part of the seating arrangement in a theatre is shown in the diagram below. The seats are arranged in rows. Row 1 is nearest the stage and has 28 seats. Each subsequent row behind that contains one more seat. i.e. row 2 has 29 seats, row 3 has 30 seats, and so on.

(a) Find the number of seats in row 10.



(b) There are 50 seats in the last row. How many rows of seats are there in the theatre?

(c) Find the total number of seats in the theatre.

(d) On a particular night 600 people are at a show in the theatre. Assume that people are seated only in the rows closest to the stage, i.e. they have filled the first n rows and there are some people seated in the next row.
Find the value of n **and** find the number of people seated in the next row.

Value of n = Number of people in next row =

(e) For a particular show, adult tickets cost €25 each and children's tickets cost €12 each.
Find the total income from ticket sales if 276 adult tickets and 212 children's tickets were sold.

(f) **(i)** For a different show 752 tickets were sold.
The ratio of adult tickets sold to children's tickets sold is 3 : 1.
Find how many adult tickets **and** how many children's tickets were sold for that show.

Number of adult tickets =

Number of children's tickets =

(ii) For the show referred to in part **f(i)** an adult ticket cost $2\frac{1}{2}$ times as much as a children's ticket. The income was €17 578. Find the cost of a children's ticket for this show.

Cost of a children's ticket =

Section B		
Q7	**Model Solution – 55 Marks**	**Marking Notes**
(a)	37	**Scale 5B (0, 3, 5)** *Partial Credit:* • correct number of seats for any row between row 4 and row 9 (inclusive) **Note:** Accept 37 without work
(b)	$28 + (n - 1)1 = 50$ $27 + n = 50$ $n = 23$ **Or** 50 – 37 = 13 13 + 10 = 23	**Scale 10C (0, 3, 8, 10)** *Low Partial Credit:* • any evidence of counting (e.g. listing 28, 29,…) • 13 or 22 without work • writes T_n = a + (n-1)d • a or d identified *High Partial Credit:* • 50 – 28 or 50 – 37 • $28 + (n - 1)1 = 50$ **Note:** Accept 23 without work
(c)	$S_{23} = \frac{23}{2}[2(28) + (22)(1)]$ $S_{23} = 897$ (seats) **Or** 28 + 29 + . . . + 50 = 897	**Scale 10C (0, 3, 8, 10)** *Low Partial Credit:* • S_n formula written • any evidence of counting (e.g. listing 28 + 29 + 30 +…) • a or d identified *High Partial Credit:* • full listing with addition indicated but incorrect solution • S_n formula fully substituted **Note:** Accept 897 without work

(d)	$\frac{n}{2}(56+n-1)=600$ $n^2+55n-1200=0$ $n=16{\cdot}7$ $S_{16}=568$ $600-568=32$ people in next row **Or** [28 + 29 + . . .+ 43] = 568 600 – 568 = 32 16 rows and 32 in next **Or** ([28 + 29 + . . .+ 44] = 612 612 – 44 = 568 600 – 568 = 32 (17 –1 = 16) rows and 32 in next	**Scale 10C (0, 3, 8, 10)** *Low Partial Credit:* • S_{n} formula written • a or d identified • any evidence of counting (e.g. listing 28 + 29 + 30 +…) *High Partial Credit*: • $n^2+55n-1200=0$ • $\frac{n}{2}(56+n-1)=600$ • full listing to 16 or 17 terms with addition indicated but incorrect solution **Note:** Accept (16, 32) without work
(e)	$276(25)+212(12)$ 6900 + 2544 = €9 444	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* • $276(25)$ • $212(12)$ *High Partial Credit*: • $276(25)+212(12)$ formulated **Note**: Accept correct answer without work
(f) **(i)**	$\frac{752}{4}$ = 188 Children's tickets $188\times 3=564$ Adult tickets	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* • 4 • $\frac{752}{x},\ x\neq 4,1$ *High Partial Credit:* • 188 found or 564 **Note**: Accept correct answer without work
(f) **(ii)**	$188x+564(2{\cdot}5x)=17\,578$ $1598x=17\,578$ $x=€11$	**Scale 10C (0, 3, 8, 10)** *Low Partial Credit:* • $x,\ 2{\cdot}5x$ • correct answer without work *High Partial Credit*: • $188x+564(2{\cdot}5x)=17\,578$, or consistent with f(i)

The general term of an arithmetic sequence is $T_n = 15 - 2n$, where $n \in \mathbb{N}$.

(a) **(i)** Write down the first three terms of the sequence.

(ii) Find the first negative term of the sequence.

(b) **(i)** Find $S_n = T_1 + T_2 + \cdots + T_n$, the sum of the first n terms of the series, in terms of n.

(ii) Find the value of n for which the sum of the first n terms of the series is 0.

(a) **(i)** Write down the first three terms of the sequence.

$T_1 = 15 - 2(1)$ $\quad$ $T_2 = 15 - 2(2)$ $\quad$ $T_3 = 15 - 2(3)$

$T_1 = 13$ $\quad$ $T_2 = 11$ $\quad$ $T_3 = 9$

Note: Accept $T_0,\ T_1,\ T_2$ $\quad$ 15, 13, 11

(ii) Find the first negative term of the sequence.

$15 - 2n < 0$

$-2n < -15$

$2n > 15$ $\quad$ or $\quad$ -1

$n > 7\frac{1}{2}$

=> The 8th term

(b) **(i)** Find $S_n = T_1 + T_2 + \cdots + T_n$, the sum of the first n terms of the series, in terms of n.

$$S_n = \frac{n}{2}\left(2(13) + (n-1)(-2)\right)$$

(ii) Find the value of n for which the sum of the first n terms of the series is 0.

$$S_n = \frac{n}{2}(-2n + 28) = [-n^2 + 14n]$$

$$\frac{n}{2}(-2n + 28) = 0$$

$$n(-2n + 28) = 0$$

$$-2n + 28 = 0$$

$$n = 14$$